DeepShovel 2.0: A Scientific Reasoning Network for Evidence-Grounded Verification and Discovery

Haoxiang Zhang
1227538797@sjtu.edu.cn
Shanghai Jiao Tong University
Shanghai, China

Jiaxin Ding*
jiaxinding@sjtu.edu.cn
Shanghai Jiao Tong University
Shanghai, China

ABSTRACT

Scientific knowledge is reusable when a conclusion remains connected to the evidence, assumptions, and inferential steps that justify it. We present DeepShovel 2.0, an operational system that reconstructs the principal arguments of scientific papers and integrates them into a curated domain reasoning network. Each argument retains its jointly required premises, applicability conditions, and traceable links to source passages, data, tables, and figures. Reasoning-aware retrieval and retrieval-augmented generation assemble supporting arguments and applicable counterevidence, enabling verification of individual claims and the interdependent conclusions of one or more papers. Research agents operate on the same network to derive candidate claims, challenge their premises, and refine research strategies through successive investigations. A Codex-orchestrated geoscience study screened 48 candidates across eight rounds and produced a reproducible manuscript on regional transfer of a seawater rare-earth-element pH calibration. The demonstration connects paper understanding, premise-level verification, and agent-driven exploration through a common, inspectable scientific argument model.

CCS CONCEPTS

• **Information systems** → **Digital libraries and archives**; *Retrieval models and ranking*; • **Computing methodologies** → **Knowledge representation and reasoning**.

KEYWORDS

scientific reasoning networks, argument reconstruction, evidence provenance, claim verification, retrieval-augmented generation, research agents

1 INTRODUCTION

Scientific conclusions depend on observations, assumptions, and inferential steps. Reuse requires recovering why they hold and where they apply: a calibration's original fit, for example, does not establish its transferability.

DeepShovel 2.0 makes the *source-grounded scientific argument* the reusable unit of knowledge. It reconstructs each admitted paper's research question, main claims, and inferential dependencies, then connects these arguments across selected domain literature. The network preserves the conditions under which conclusions support, qualify, or challenge one another. It supports two complementary operations: tracing a conclusion backward to its justification, and extending grounded premises forward into new, testable inferences.

For the scholarly Web, publications, figures, tables, and data become addressable components of cross-document reasoning. Source links connect an answer to an inference and its original evidence. The running system is at https://deepshovel.iiot-shxt.xyz. Code, fixed cases, an English walkthrough, and the research reproduction package are released at https://github.com/spiderLL2002/deepshovel, tag `www2027-artifact-v1.1.1`. A service snapshot on 27 October 2026 reported 9,506 active paper families and 41,092 active claims. Expert review of development questions and datasets and additional spot checks support quality control; these activity counts do not imply individually expert-certified inferences.

Contribution and positioning. GeoDeepShovel supports scientific database construction [1]; SciFact formalizes claim verification with evidence rationales [2]. Argument Web benchmarks prediction of argumentative relations across discourse units [3]; GraphRAG supports graph-based synthesis [4]. ResearchAgent iteratively develops research ideas [5]; DeepEvidence coordinates breadth/depth exploration across biomedical knowledge sources and records an evolving evidence graph [6]. Our contribution is *continuity of scientific justification*: the joint inputs and applicability conditions of a paper's internal inferences survive retrieval, while research records connect new deductions to sources and analyses. The demonstration exposes the difference between a short source path and preserved joint dependencies, and follows retrieved arguments into research decisions.

2 COMPILING A DOMAIN REASONING NETWORK

Figure 1 shows conditions retained in the running claim inspector.

Recover the paper's scientific meaning. Ingestion parses sections, equations, references, tables, figure panels, captions, and supplementary data into addressable source objects. Reconstruction follows the paper's main reasoning routes: question → methods and assumptions → observations → intermediate interpretations → conclusions. Backward reconstruction gathers claim premises; forward reconstruction checks how observations support intermediate and final claims.

A reasoning step is represented as

$$r = (P_r, A_r, w_r, c_r, \sigma_r, \pi_r), \tag{1}$$

*Corresponding author.

WWW '27, May 10–14, 2027, Dublin, Ireland
2027.

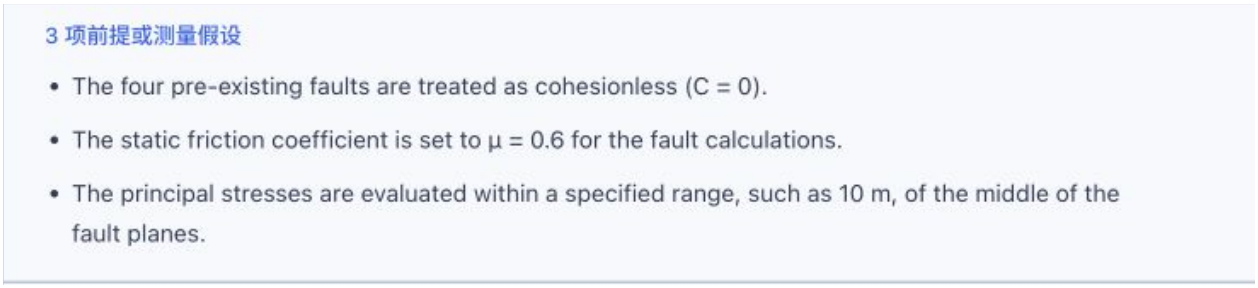


Figure 1: Actual F2 inspector with its three expanded assumptions. The release provides uncropped screenshots, the exact query, original-page inspection, and an English walkthrough.

where P_r contains jointly required premises, A_r assumptions, w_r the inferential warrant or analysis, c_r the conclusion, σ_r applicability conditions, and π_r provenance. An explicit inference node joins all co-premises. Arguments may branch through multiple observations or converge through intermediate results; one path to a conclusion can omit other required inputs.

In the stored Asmari graph [9], `reason.f2.evidence` receives two `premise_for` and three `assumption_for` inputs (Figure 1). Its downstream pressure inference also requires current-stress and critical-pressure reasoning. This explicit dependency set explains why retrieving one source path is insufficient.

Scope records material or population, setting, method, comparison, and spatial or temporal range. Reconstruction preserves quantifiers and negation and distinguishes an observation, an association, a mechanistic interpretation, and a prediction. Moving between these levels is itself an inference requiring justification. Provenance binds evidence to a document version and a passage, equation, table cell, figure panel, or dataset record. Numerical observations retain units and uncertainty; figure interpretations retain axes, captions, and experimental context. Author-stated arguments, reconstructed links, and agent-proposed inferences have distinct origins. Missing premises remain visible obligations.

Compose arguments with their conditions intact. Cross-paper alignment compares proposition meaning and scope, retaining individual claim instances and support, challenge, qualification, and dependency relations. Combining findings introduces a bridge inference with inspectable compatibility assumptions. Contradictions may target an observation, an assumption, or the warrant connecting premises to a conclusion.

Evidence lineage groups paper versions, reused datasets, and derivative analyses to distinguish repeated measurements from independent confirmation. Source fidelity, premise coverage, methodological adequacy, scope, and counterevidence inform review. Displayed confidence is a model assessment, not a calibrated truth probability.

Execution boundary. Language-model reconstruction assigns `premise_for` and `assumption_for` inputs to explicit inference nodes; these assignments distinguish required inputs from background. Separate inference nodes preserve alternative routes. Deterministic graph operations follow the recorded assignments and source identities. A warrant is a readable justification or analysis reference, not a machine-checked proof. In the released code, both interactive retrieval and final comparison guard direct contradictions and equivalences when scope is disjoint or unknown. Different free-text scientific labels alone yield unknown scope. Source review and appropriate analyses establish scientific adequacy.

3 REASONING-LEVEL RETRIEVAL AND VERIFICATION

Retrieve the justification of an answer. Requests identify a target proposition, scope, and evidential needs. Lexical/semantic retrieval recalls claims and observations; reranking prioritizes relevance and applicability. Support and counterevidence searches cover incompatible observations, alternative mechanisms, failed transfers, and challenged assumptions.

The search list offers a short preview. Opening a claim executes backward dependency expansion, following all represented premise and assumption inputs and their grounding links. The detail response retains the resulting *argument bundle* and marks `truncated` if its depth bound (default 32) stops expansion. This checks completeness relative to recorded dependencies; it cannot detect scientific premises never extracted from the source. Synthesis budgets and scientific sufficiency are separate from this explicit traversal bound. The released comparison adapter carries typed dependencies, assumption roles, and completeness to both model stages. Selecting evidence cannot discard this dependency context; explicitly truncated inputs require an unresolved result.

RAG synthesizes a scoped answer with assertions linked to reasoning steps and evidence. Retrieval prioritizes premises needed to justify a proposition or change its interpretation.

Locate the disagreement inside the argument. Verification distinguishes support, contradiction, qualification, and unresolved evidence. A counterexample must address the claimed scope; another regime may instead challenge a generalization. The challenged observation, assumption, or inference locates the disagreement. Dependency traversal connects a paper's principal claims; alignment across papers exposes differences in data, methods, or applicability. Alternative support can preserve a conclusion when one route fails.

Executable checks of recorded premises. The release executes the unchanged production detail expander on all seven claims in the

saved Asmari graph [9]. An independent fixed-point traversal supplies the reference edge sets: all seven full expansions match. For F2, the search preview retains 1/18 recorded premise edges and 1/3 assumption edges; the detail expansion retains **18/18 and 3/3**, in 41 entities and 37 edges. Thus users can recover the three-method agreement, current-stress reasoning, and three assumptions omitted by a short trace. This directly checks joint-input preservation, beyond path existence.

Depth limits 1, 2, and 3 produce 21 bounded runs, all marked truncated; all 19 runs omitting premise or assumption edges are flagged. Three controlled missing-input mutations are rejected by the production validator. A retained negative example also shows that an unrecorded internal dependency can pass structural validation. These are checks on one paper, not extraction-accuracy estimates. A cross-language fixture additionally checks F2 through the released comparison projection, Python schema, and both model-input stages, with controlled model replies. It checks structure transport and truncation guards, not model accuracy.

The live walkthrough opens F2's original page 11, reasoning steps, and assumptions (Figure 1). Source inspection qualifies the imported scalar 55 as the approximate 55–57 MPa maximum sustainable pore pressure under zero cohesion and friction 0.6, rather than a pressure increment or statistical interval. A known wrong-column rectangle remains documented; page and quotation inspection are required. The release also preserves three Nd queries where the same candidate changes from related to supporting to related as the target changes. These are inspectable model decisions; synthetic scope-guard fixtures are labeled separately.

4 AGENTS FOR DISCOVERY AND RECURSIVE IMPROVEMENT

Agents retrieve a target's justification, identify an unresolved transition, propose a conditional inference, and seek counterexamples and prior work. Candidates retain their derivation, bridging assumptions, and discriminating analysis. Task-level recursive self-improvement (RSI) records feedback-driven revisions to the investigation procedure.

An executed geoscience investigation. A Codex-orchestrated run screened 48 candidates over eight rounds in rare-earth geochemistry and ocean anoxic-event research, advancing one to a reproducible manuscript. In a documented screening decision, DeepShovel returned Hongo et al.'s [10] scoped REE–Mg mixing argument. The agent followed its source chain to the original methods and Figures 10–12, established prior overlap, and retained the filtered/unfiltered sampling distinction. The candidate record explicitly cites this source as strengthening the rejection of REE-R7-02. Released request timestamps, returned argument identifiers, PDF hash, and decision excerpts expose that contribution to research selection. The surviving question tested regional transfer of a published seawater REE–pH calibration [7].

The run combined DeepShovel retrieval and claim comparison with original publications and public GEOTRACES GP15 observations [8]. REE and pH were matched by parent-bottle identity, with distinct filtered/unfiltered analytical aliquots. The target, calibration, and depth reference use total-scale pH at 25°C and surface pressure; no basis conversion was needed. The published chemical

Table 1: Recorded research outcome: RMSE in pH units across six equal-weight layer means. Both methods use the same eligible samples within each layer. These errors describe the geochemical comparison.

GP15 group	Bottles / stations	REE model	Depth reference
Leg 1	24 / 6	0.070	0.170
Leg 2	55 / 8	0.218	0.110

predictor and its original depth reference were fixed before comparison. Across six 100-m layers between 200 and 800 m, their error ordering reversed between the two station groups (Table 1). This produced a precise claim: retaining the original depth interval did not ensure improvement over the original reference on the sampled support. Figure 2 makes the joint premises and the challenged generalization explicit.

The research record also contains a deductive extension. The Leg 2 layerwise squared-error differences $d_i = e^2_{\mathrm{REE},i} - e^2_{\mathrm{ref},i}$, in ascending depth order, are 0.07285, 0.05368, 0.05320, 0.02076, 0.00694, and 0.00545 pH^2 (rounded). Hence $\sum_i w_i d_i > 0$ for any $w_i \geq 0$, $\sum_i w_i = 1$: reweighting these same layer means cannot reverse the reference's advantage. The final claim–source map stores this as C12, binds it to the hashed layer-difference table, and restricts it to unchanged within-layer support. C03 separately binds the empirical RMSEs to their aggregate and layer records. Codex persists this claim ledger and its source maps in the research workspace. The propositions, with their scope, are sent as text to DeepShovel for itemwise comparison; its returned assessments are stored alongside the local analysis. A released lineage index joins the source-graph PDF hash to the research source-input file, frozen protocol, and C01/C03/C12 file hashes. This index was assembled from existing records for publication; graph and research-ledger identifiers remain separate, and automatic reinsertion is not asserted.

The run also records a procedure revision. A pre-execution review checked revised uncertainty rules requiring at least six station labels and 90% complete-support resamples, and suppressing aggregate comparisons when station deletion removes an originally represented layer. The frozen protocol binds that review; the subsequent run applies the rules. The released record links the review, frozen conditions, and executed outputs, making this task-level RSI episode inspectable without attributing authorship of the rules or claiming a measured improvement.

The public package includes exact model coefficients, depth-reference values, sample joins, the frozen protocol, manuscript, and claim maps. A fresh ten-stage run regenerated 76 reference files byte-identically. Sensitivities preserve the error ordering under four aggregation choices, station deletion, and alternate treatment of one unresolved depth. DeepShovel's seven final claim comparisons reported incomplete literature coverage and accepted no comparable literature relation. Codex-led source inspection established model definitions and sample identity; deterministic analysis supplied numerical support for the local claim. These distinct evidence channels remain labeled. The episode demonstrates completed research and procedure refinement; it does not isolate a performance gain from RSI.

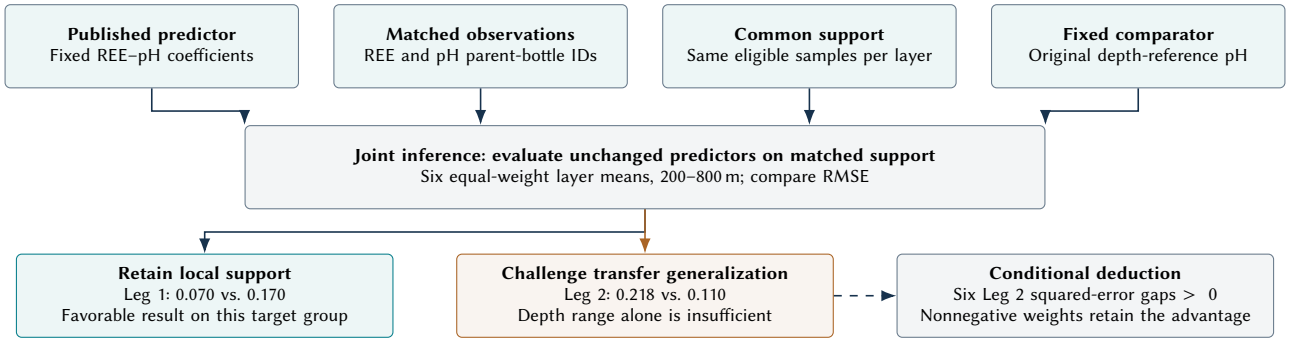


Figure 2: Instantiated reasoning structure for the completed GP15 investigation. The four top-row premises jointly warrant the comparison. The adverse branch challenges a generalization posed for verification, while the source's original calibration and the favorable local result remain available. The final box holds for the same unperturbed layer means; changing within-layer support requires a new analysis.

5 INTERACTIVE DEMONSTRATION

Attendees trace a published argument to a tested claim, controlling the comparison scope and next research question.

1. Inspect a paper's reasoning (live). Paste the released F2 query into the Web workbench. Open the representative paper, inspect the 55–57 MPa claim, expand its observations, reasoning steps and three assumptions, and open original page 11. Compare the short search preview with the complete recorded dependency set. The English guide translates the current Chinese controls and links each action to a real screenshot.

2. Test scope and completeness (live and replay). Change the query's setting or request counterevidence in the live service. Inspect the retained scope and relation explanations. Then replay the fixed Nd query trio and execute the released Asmari expansion with a lower depth bound: the response explicitly reports truncation. Full dependencies, short previews, and controlled invalid inputs are separately available for inspection.

3. Follow evidence into research (saved and executable). Open the Hongo retrieval-to-decision trace, then follow the GP15 source hash into the frozen protocol and C03/C12 outputs. Recompute the numerical claims locally; inspect the procedure revision and resulting manuscript. Attendees can request new counterevidence or propose an independent station group through the connected agent. New scientific investigations are not represented as instant replay results.

Venue operation and reproducibility. At the venue, the shared Web service provides live source inspection; the laptop runs the tagged release's comparison-input and scope-guard checks on recorded inputs with controlled model replies. These release fixes are not deployed to the shared service. New retrieval and agent calls require connectivity and configured services. The tagged repository supplies application source, fixed graphs, licensed sources, interface captures, English instructions, and locked dependencies. Dependency and research replays need no model key; saved model outputs remain labeled. Fixed materials remain usable offline after setup, including SHA-256-verified restoration of two supporting inputs from their providers.

6 ETHICAL USE OF DATA

Publications and research data are used subject to their licenses and access permissions. Attribution, source versions, and derived-data provenance are retained. Expert-reviewed questions and datasets and spot checks support quality control; generated hypotheses retain their provisional status. The example reanalyzes public geochemical measurements and includes no human-subject experiment. Assessments concern scientific claims and applicability, not researchers' integrity.

REFERENCES

[1] Shao Zhang et al. 2023. GeoDeepShovel: A platform for building scientific database from geoscience literature with AI assistance. *Geoscience Data Journal* 10, 4, 519–537. https://doi.org/10.1002/gdj3.186
[2] David Wadden et al. 2020. Fact or Fiction: Verifying Scientific Claims. In *EMNLP*, 7534–7550. https://doi.org/10.18653/v1/2020.emnlp-main.609
[3] Florian Ruosch, Cristina Sarasua, and Abraham Bernstein. 2025. Mining Inter-Document Argument Structures in Scientific Papers for an Argument Web. *Transactions on Graph Data and Knowledge* 3, 3, Article 4. https://doi.org/10.4230/TGDK.3.3.4
[4] Darren Edge et al. 2024. From Local to Global: A Graph RAG Approach to Query-Focused Summarization. *arXiv:2404.16130*. https://arxiv.org/abs/2404.16130
[5] Jinheon Baek, Sujay Kumar Jauhar, Silviu Cucerzan, and Sung Ju Hwang. 2025. ResearchAgent: Iterative Research Idea Generation over Scientific Literature with Large Language Models. In *NAACL*, 6709–6738. https://doi.org/10.18653/v1/2025.naacl-long.342
[6] Zifeng Wang et al. 2026. Empowering biomedical evidence exploration and synthesis with deep knowledge graph research. *Nature Machine Intelligence* 8, 1142–1156. https://doi.org/10.1038/s42256-026-01266-0
[7] Ping-Chun Lin and David C. Catling. 2024. The Rare Earth Element Distribution in Marine Carbonates as a Potential Proxy for Seawater pH on Early Earth. *American Journal of Science* 324, Article 10. https://doi.org/10.2475/001c.118215
[8] BCO-DMO. GEOTRACES GP15 rare earth element and pH datasets. REE: https://doi.org/10.26008/1912/bco-dmo.932161.1, https://doi.org/10.26008/1912/bco-dmo.932559.1; pH: https://doi.org/10.26008/1912/bco-dmo.838157.2, https://doi.org/10.26008/1912/bco-dmo.838173.2.
[9] Majid Taghipour et al. 2019. Estimation of the current stress field and fault reactivation analysis in the Asmari reservoir, SW Iran. *Petroleum Science* 16, 513–526. https://doi.org/10.1007/s12182-019-0331-9
[10] Yayoi Hongo et al. 2007. Rare Earth Elements in the hydrothermal system at Okinawa Trough back-arc basin. *Geochemical Journal* 41, 1, 1–15. https://doi.org/10.2343/geochemj.41.1